

EEG-Based Seizure Detection Using Logistic Regression: A Comparative Study of Feature Selection, PCA, Regularisation, and Class-Imbalance Handling

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Abstract—Electroencephalography (EEG) is widely used for studying brain activity and detecting epileptic seizure events. This report presents a comparative machine-learning study for EEG-based seizure detection using three datasets: Bonn EEG, Epileptic Seizure Recognition, and BEED. Two preprocessing pipelines were evaluated: standardisation followed by ANOVA feature selection, and Min-Max scaling followed by principal component analysis (PCA). Logistic Regression was used as the baseline classifier. The study further analysed the effect of regularisation methods, including L1, L2, and Elastic-Net, and class-imbalance handling techniques such as SMOTE, under-sampling, and balanced class weights. Results show that Pipeline-1 generally performed better than Pipeline-2, especially on the Bonn EEG and Seizure datasets. BEED achieved the highest F1-score, while the Seizure dataset remained challenging due to severe class imbalance and low minority-class recall. The findings highlight that preprocessing, regularisation, and imbalance handling significantly influence seizure-detection performance.

Index Terms—EEG, seizure detection, logistic regression, machine learning, ANOVA feature selection, PCA, SMOTE, class imbalance, regularisation

I. INTRODUCTION

Epileptic seizure detection is an important biomedical signal-processing problem because seizures are often reflected in abnormal electrical brain activity recorded through electroencephalography (EEG). Manual interpretation of EEG signals requires expert knowledge and can be time-consuming. Machine-learning methods can assist by learning discriminative patterns from EEG-derived features and classifying whether a signal segment corresponds to seizure or non-seizure activity.

This report investigates EEG-based seizure detection using Logistic Regression across three EEG-related datasets. The study focuses not only on classification accuracy but also on F1-score and precision-recall area under the curve (PR-AUC), which are more informative when datasets are imbalanced. The main objectives are:

- to compare two preprocessing pipelines for EEG classification;
- to evaluate Logistic Regression performance across multiple EEG datasets;

- to study bias-variance behaviour using different regularisation strengths;
- to compare L1, L2, and Elastic-Net regularisation;
- to analyse the effect of class-imbalance handling methods.

II. RELATED WORK

EEG-based seizure detection has been widely studied using signal-processing and machine-learning techniques. Traditional machine-learning approaches often rely on feature extraction, dimensionality reduction, and supervised classifiers. Logistic Regression remains useful as a baseline model because it is interpretable, efficient, and suitable for binary classification. However, EEG datasets can have high dimensionality and class imbalance, making preprocessing and evaluation metric selection critical. Dimensionality-reduction methods such as PCA and feature-selection methods such as ANOVA F-score are commonly used to reduce noise and improve computational efficiency.

III. DATASETS

Three EEG-related datasets were used in this study. Table I summarises their size, number of features, and class distribution.

TABLE I
DATASET SUMMARY

Dataset	Rows	Features	Class Distribution
Bonn EEG	500	4097	0: 400, 1: 100
Seizure	11500	178	0: 9200, 1: 2300
BEED	8000	16	0: 2000, 1: 6000

The class-distribution analysis showed imbalance in all three datasets. Bonn EEG and Seizure had fewer seizure/event samples, while BEED had more event-class samples than non-event samples. This justified the use of F1-score, recall, precision, and PR-AUC in addition to accuracy.

IV. METHODOLOGY

A. Preprocessing Pipeline 1

Pipeline-1 consisted of two steps:

- 1) Standardisation using zero mean and unit variance.
- 2) ANOVA F-score feature selection using SelectKBest.

For Bonn EEG and Seizure, the top 50 features were selected. For BEED, all 16 available features were retained. The final output shapes were:

- Bonn EEG: 500 samples, 50 features;
- Seizure: 11500 samples, 50 features;
- BEED: 8000 samples, 16 features.

B. Preprocessing Pipeline 2

Pipeline-2 consisted of:

- 1) Min-Max scaling to transform features into the range $[0, 1]$.
- 2) Principal Component Analysis for dimensionality reduction.

For Bonn EEG and Seizure, 30 principal components were retained. For BEED, 16 components were used because only 16 original features were available. PCA retained 70.2% variance for Bonn EEG, 86.5% for Seizure, and 100% for BEED.

C. Classifier

Logistic Regression was used as the classification model. The dataset was split into training and testing sets using an 80:20 stratified split with a fixed random state. The model was trained using the SAGA solver.

D. Evaluation Metrics

The following metrics were used:

- Accuracy;
- F1-score;
- Precision;
- Recall;
- PR-AUC.

F1-score and PR-AUC were emphasised because seizure-detection datasets are often imbalanced, and accuracy alone may give a misleading impression of performance.

V. RESULTS AND DISCUSSION

A. Baseline Pipeline Comparison

Table II shows the baseline Logistic Regression results for both preprocessing pipelines.

Pipeline-1 outperformed Pipeline-2 on Bonn EEG and Seizure. This suggests that ANOVA-based feature selection was more useful than PCA for retaining discriminative EEG features in these datasets. BEED showed similar performance across both pipelines, likely because it had only 16 original features and did not require aggressive dimensionality reduction.

The Seizure dataset showed high accuracy but very low F1-score. This indicates that the model was biased toward the majority class and had difficulty correctly detecting the minority seizure/event class.

TABLE II
BASELINE LOGISTIC REGRESSION RESULTS

Dataset	Pipeline	Accuracy	F1	PR-AUC
Bonn EEG	Pipeline-1	0.9100	0.7273	0.7922
Bonn EEG	Pipeline-2	0.8800	0.5714	0.6276
Seizure	Pipeline-1	0.8113	0.1107	0.5048
Seizure	Pipeline-2	0.8039	0.0425	0.4673
BEED	Pipeline-1	0.8006	0.8825	0.7598
BEED	Pipeline-2	0.7969	0.8807	0.7395

B. Regularisation Comparison

Table III compares L1, L2, and Elastic-Net regularisation using Pipeline-1.

TABLE III
REGULARISATION COMPARISON

Dataset	Regularisation	F1	PR-AUC	Zero Coeffs. (%)
Bonn	L1	0.7500	0.7585	48.0
Bonn	L2	0.7273	0.7922	0.0
Bonn	Elastic-Net	0.7273	0.7853	30.0
Seizure	L1	0.0990	0.5056	16.0
Seizure	L2	0.1107	0.5048	0.0
Seizure	Elastic-Net	0.0990	0.5054	4.0
BEED	L1	0.8828	0.7585	0.0
BEED	L2	0.8825	0.7598	0.0
BEED	Elastic-Net	0.8825	0.7591	0.0

L1 regularisation produced sparse coefficients, especially for Bonn EEG, where 48% of the coefficients became zero. This indicates that L1 can perform embedded feature selection. L2 regularisation gave stable performance, while Elastic-Net provided a compromise between sparsity and stability. The BEED dataset showed almost identical F1-score across all regularisation methods.

C. Class-Imbalance Handling

Table IV summarises the results of imbalance-handling techniques.

For the Seizure dataset, imbalance handling greatly improved recall and F1-score compared with no adjustment. However, precision decreased, showing a trade-off between detecting more seizure cases and increasing false positives. For Bonn EEG, balanced weights produced the best F1-score. For BEED, the unadjusted model produced the highest F1-score because the event class was the majority class, but imbalance methods improved PR-AUC.

D. Bias-Variance Analysis

The bias-variance analysis was performed by varying the regularisation strength C . A very small value of C applies strong regularisation and can lead to underfitting. A very large value weakens regularisation and may increase overfitting risk. The learning-curve analysis helped identify whether the model was limited by bias, variance, or data imbalance. The results suggested that performance on the Seizure dataset was limited

TABLE IV
CLASS-IMBALANCE HANDLING RESULTS

Dataset	Method	F1	Precision	Recall	PR-AUC
Bonn	No Adjustment	0.7273	0.9231	0.6000	0.7922
Bonn	SMOTE	0.7179	0.7368	0.7000	0.7452
Bonn	Under-Sampling	0.7000	0.7000	0.7000	0.7934
Bonn	Balanced Weights	0.7368	0.7778	0.7000	0.7715
Seizure	No Adjustment	0.1107	0.9643	0.0587	0.5048
Seizure	SMOTE	0.3908	0.3140	0.5174	0.4865
Seizure	Under-Sampling	0.3783	0.3068	0.4935	0.4810
Seizure	Balanced Weights	0.3849	0.3137	0.4978	0.4764
BEED	No Adjustment	0.8825	0.7908	0.9983	0.7598
BEED	SMOTE	0.7377	0.8309	0.6633	0.8043
BEED	Under-Sampling	0.7202	0.8303	0.6358	0.8090
BEED	Balanced Weights	0.7337	0.8312	0.6567	0.8092

more by class imbalance and minority-class detection difficulty than by simple model complexity alone.

VI. COMPARATIVE ANALYSIS

This section presents the most important comparative interpretation of the experimental results. A strong EEG seizure-detection report should not only present accuracy values, but should also explain why the results changed across preprocessing pipelines, regularisation methods, and imbalance-handling strategies.

A. Does Preprocessing Order Affect Results?

Yes, preprocessing order affected the results clearly. Pipeline-1, which used standardisation followed by ANOVA feature selection, generally performed better than Pipeline-2, which used Min-Max scaling followed by PCA. For the Bonn EEG dataset, Pipeline-1 achieved an F1-score of 0.7273, while Pipeline-2 achieved an F1-score of 0.5714. Similarly, for the Seizure dataset, Pipeline-1 achieved an F1-score of 0.1107, whereas Pipeline-2 achieved only 0.0425.

This shows that supervised feature selection was more useful than unsupervised dimensionality reduction in this experiment. ANOVA feature selection selects features according to their ability to separate classes, whereas PCA selects directions of maximum variance without considering class labels. Therefore, PCA may preserve high-variance features that are not necessarily useful for seizure classification [3], [4]. This explains why PCA did not consistently improve the Logistic Regression model.

For BEED, the difference between the two pipelines was small because the dataset had fewer features. When the number of features is already limited, dimensionality reduction has less effect. Therefore, preprocessing order matters more in high-dimensional EEG datasets such as Bonn EEG.

B. Which Regularisation Generalises Best Across Datasets?

Among L1, L2, and Elastic-Net regularisation, L2 regularisation appeared to generalise most consistently across datasets. L2 does not remove features completely; instead, it reduces the size of coefficients and helps control overfitting. This is useful

in EEG data because several features may contribute small but meaningful information.

L1 regularisation was useful for producing sparse models. It removed weaker features by setting some coefficients to zero. This makes the model easier to interpret, especially in high-dimensional datasets. However, L1 did not always give the best performance across all datasets.

Elastic-Net combines L1 and L2 penalties. It can be useful when there are many correlated features, but in this experiment it did not clearly dominate the other two methods. Therefore, L2 can be considered the most stable regularisation method overall, while L1 is better when feature sparsity and interpretability are important [6].

C. Does Elastic-Net Consistently Outperform L1 and L2?

No, Elastic-Net did not consistently outperform L1 and L2. Although Elastic-Net is theoretically attractive because it combines the advantages of both L1 and L2 regularisation, its performance depends on the dataset structure.

In some datasets, L1 performed slightly better because it removed unnecessary features. In other cases, L2 performed better because it retained all features while reducing overfitting. Elastic-Net gave balanced results, but it was not always the best.

Therefore, Elastic-Net should not be considered automatically superior. It is a compromise method, not a guaranteed improvement. The best regularisation method depends on feature dimensionality, feature correlation, class imbalance, and the nature of the EEG dataset.

D. How Does Imbalance Handling Interact with Regularisation?

Class imbalance had a strong effect on model performance, especially in the Seizure dataset. Without imbalance handling, the model achieved reasonable accuracy but poor recall and F1-score. This means the model was biased toward the majority class and failed to detect many seizure cases.

When imbalance-handling methods such as SMOTE, under-sampling, and balanced class weights were applied, recall improved. This means the model became better at identifying seizure cases. However, precision decreased in some cases,

meaning that the model also produced more false positives. This is a common trade-off in imbalanced classification problems [5], [7].

This shows an important interaction between imbalance handling and regularisation. Regularisation controls model complexity, while imbalance handling changes the effective class distribution during training. If the dataset remains imbalanced, even a well-regularised model may still favour the majority class. On the other hand, if imbalance correction is too strong, the model may become overly sensitive to the minority class and increase false-positive predictions.

For the Bonn EEG dataset, balanced class weights gave strong results because moderate correction was enough. For the Seizure dataset, stronger imbalance-handling methods improved recall and F1-score. For BEED, imbalance handling did not always improve F1-score because the positive class was already dominant. Therefore, imbalance handling should be selected according to the class distribution and the clinical aim of the model.

In seizure detection, recall is especially important because missing a seizure case may be more serious than producing a false alarm. Therefore, F1-score, recall, and PR-AUC are more meaningful than accuracy alone.

E. Overall Comparative Findings

The main findings of the comparative analysis are:

- Preprocessing order affected model performance.
- Standardisation followed by ANOVA feature selection generally performed better than Min-Max scaling followed by PCA.
- L2 regularisation generalised most consistently across datasets.
- L1 regularisation was useful for sparsity and interpretability.
- Elastic-Net did not consistently outperform L1 or L2.
- Imbalance handling improved recall and F1-score in highly imbalanced datasets.
- Imbalance handling may reduce precision because the model detects more minority-class cases.
- Accuracy alone was not sufficient for evaluating seizure detection.
- F1-score, recall, and PR-AUC were more reliable metrics for imbalanced EEG classification.

Overall, the results show that EEG seizure detection depends on the complete machine-learning pipeline, not only on the classifier. A simple Logistic Regression model can perform effectively when preprocessing, regularisation, and class-imbalance handling are properly selected.

VII. CONCLUSION

This report presented a comparative study of EEG-based seizure detection using Logistic Regression. Pipeline-1, based on standardisation and ANOVA feature selection, generally performed better than Pipeline-2, based on Min-Max scaling and PCA. BEED achieved the strongest F1-score, while the

Seizure dataset remained the most challenging due to severe class imbalance and poor minority-class recall.

Regularisation analysis showed that L1 can create sparse models, while L2 provides stable performance. Elastic-Net offered a balanced approach but did not consistently outperform L1 or L2. Class-imbalance handling improved recall and F1-score for the Seizure dataset, although at the cost of precision.

Overall, the study demonstrates that preprocessing and class-imbalance handling are critical in EEG seizure detection. Future work may include more advanced classifiers such as Support Vector Machines, Random Forests, XGBoost, or deep-learning models such as CNNs and LSTMs for temporal EEG pattern learning.

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